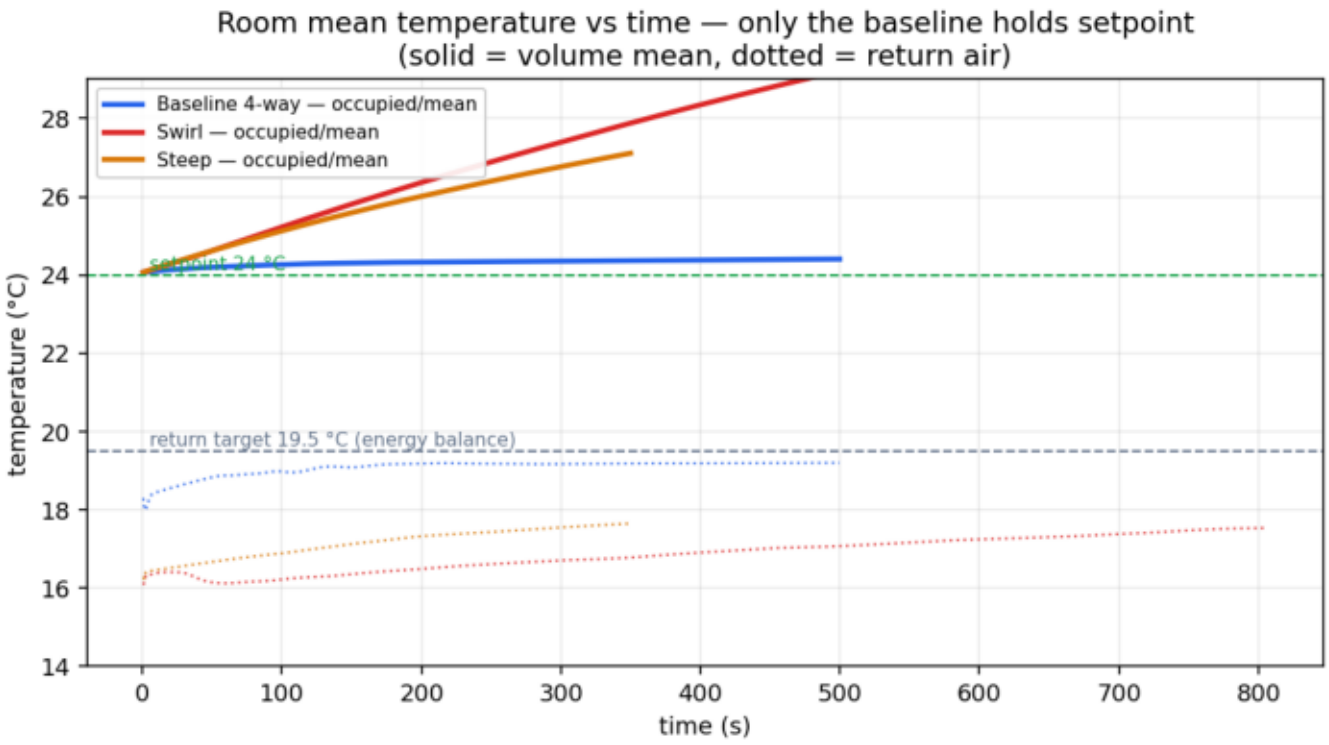
Ceiling-Cassette Diffuser Study — Setpoint Comparison

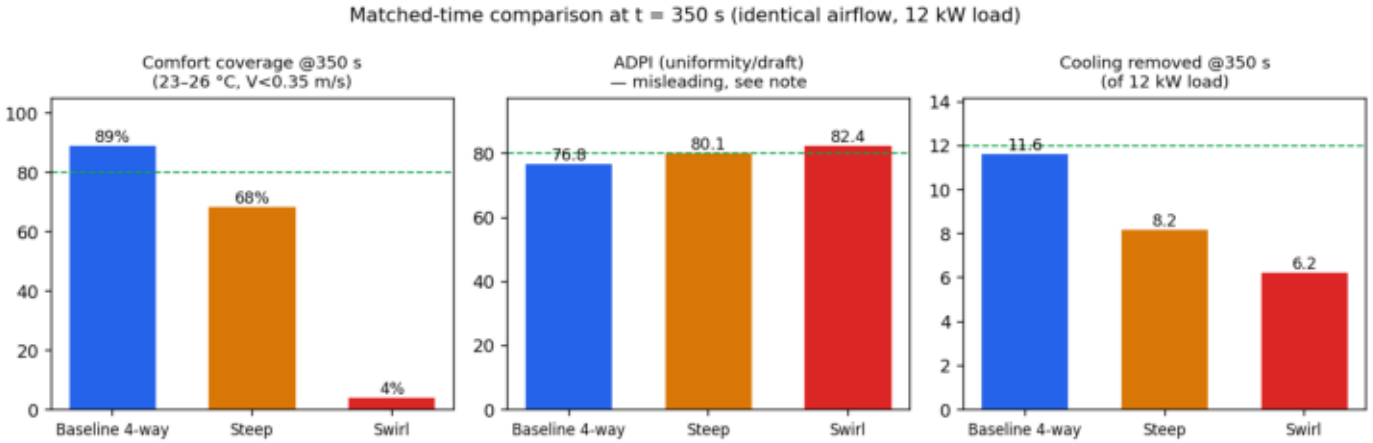
12×10×4 m hall · two 4-way cassettes · identical airflow 1.81 m³/s · 14 °C supply · 12 kW load · OpenFOAM buoyantPimpleFoa

KEY FINDING

All three patterns remove the SAME air mass (2.23 kg/s) and, by energy balance, all reach the SAME exit-air equilibrium (19.4 °C, 12 kW removed). What differs is HOW the air is used. The baseline mixes the supply int room, so its exhaust is warm (carrying heat out) and the occupied zone holds the 24 °C setpoint. Swirl and st SHORT-CIRCUIT — cold supply loops back to the central return before doing work — so at a matched 350 s th is 3–4 °C warmer and far less of the floor is comfortable, and they approach a hotter, slower equilibrium.



Matched-time comparison at t = 350 s

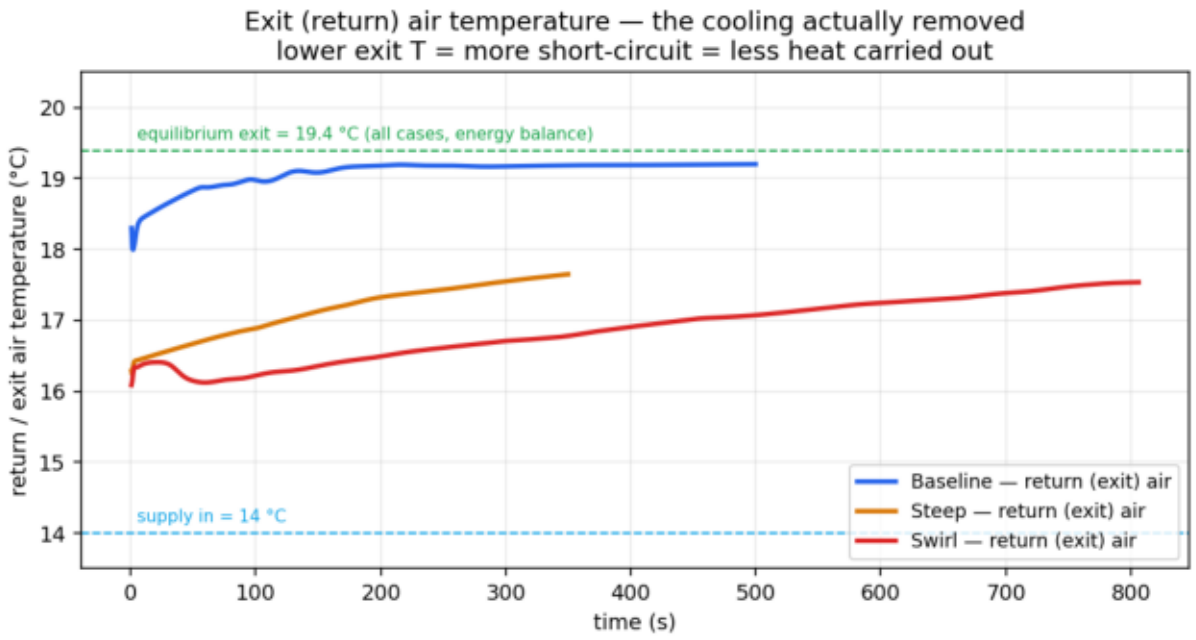


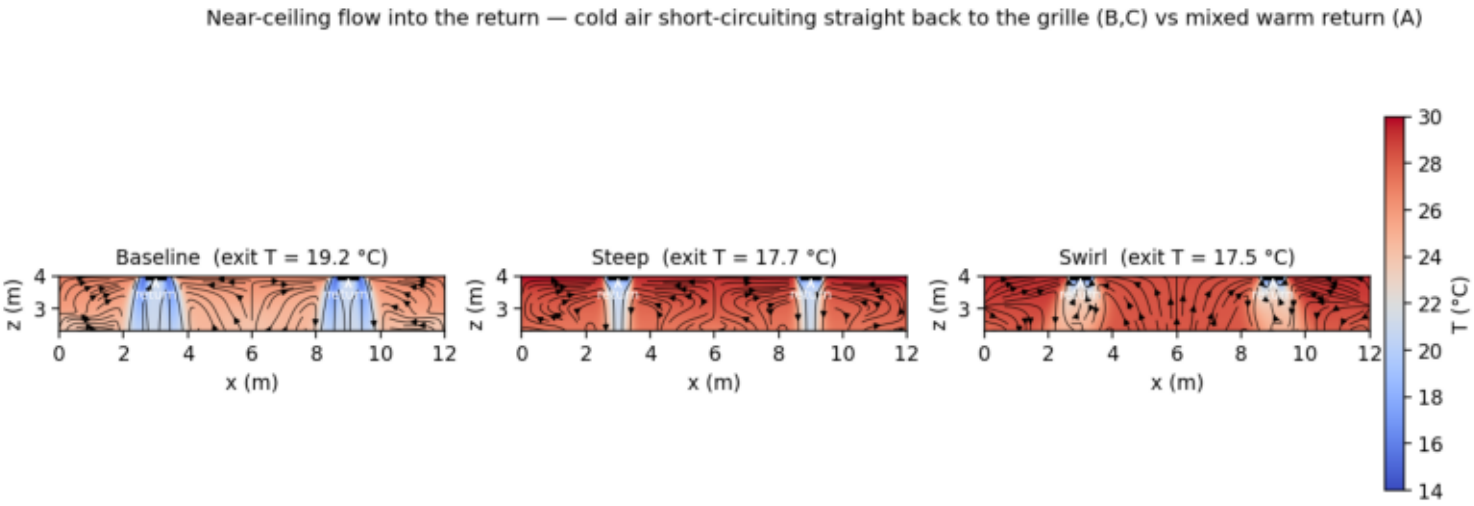
Pattern	Occ. mean	Exit/return	Cooling out	Tstd	ADPI	Coverage	Lag
Baseline 4-way	24.4 °C	19.2 °C	11.6 kW	0.50	77	89%	5.2 °C
Steep	27.1 °C	17.7 °C	8.2 kW	0.43	80	68%	9.5 °C
Swirl	27.9 °C	16.8 °C	6.2 kW	0.37	82	4%	11.1 °C

ADPI caveat:
ADPI is scored relative to the LOCAL mean, so swirl's uniformly-too-warm, dead-calm zone scores the highest ADPI (82) while giving the lowest real comfort coverage (4%). Rank on absolute coverage + cooling removed, not ADPI alon

Equilibrium note:
Energy balance fixes the exit at 14 + 12000/(2.227·1004) = 19.4 °C and cooling at 12 kW for ALL three — they only differ in how long they take and how hot the room gets on the way. Baseline is there by ~185 s; swirl is still climbing (31.5 °C at 806 s) toward a hotter, slower asymptote. These matched-350 numbers are a snapshot of that approach.

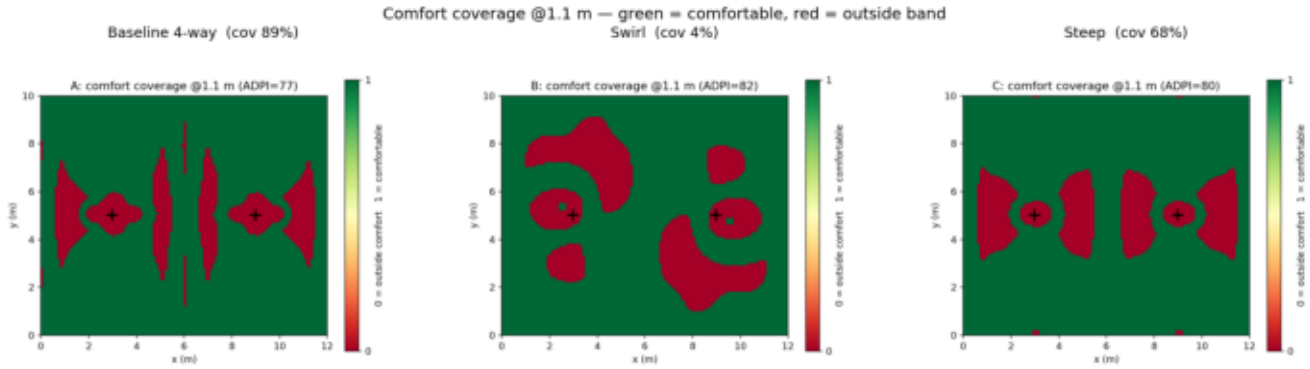
Exit (return) flow — the mechanism





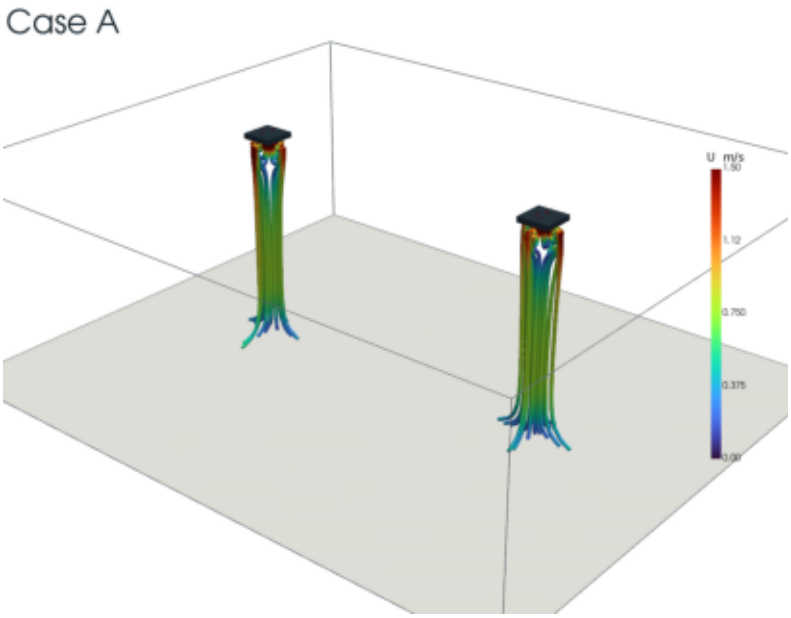
The exit-air temperature IS the cooling actually delivered: Q_removed = ṁ·cp·(T_exit − 14).
Baseline exhausts warm (19 °C → 11.6 kW removed); swirl/steep exhaust cold (17 °C → ~8 kW) because cold supply short-circuits straight back into the central return (see the blue air curling into the grille in the slices).

Comfort coverage @1.1 m (green = comfortable)

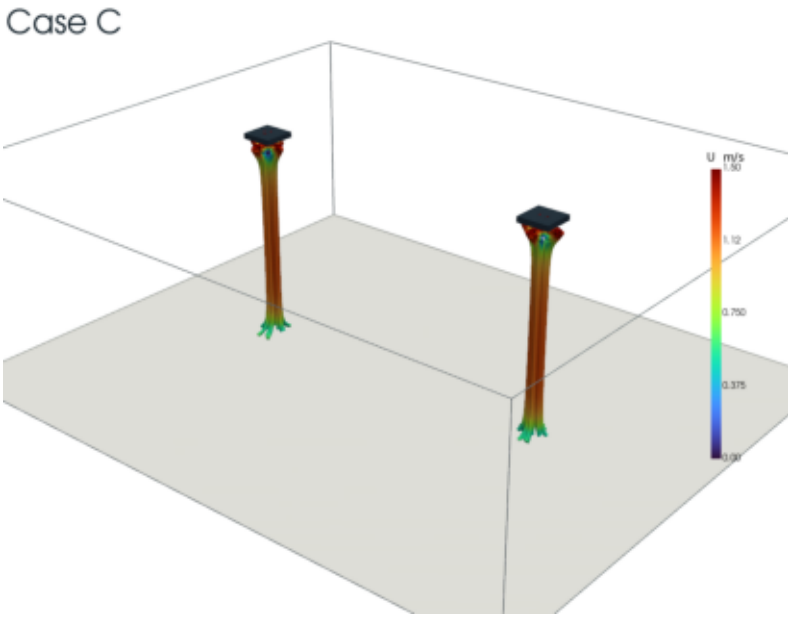


Baseline fills the plan; steep leaves gaps; swirl is almost entirely out of band at 350 s.

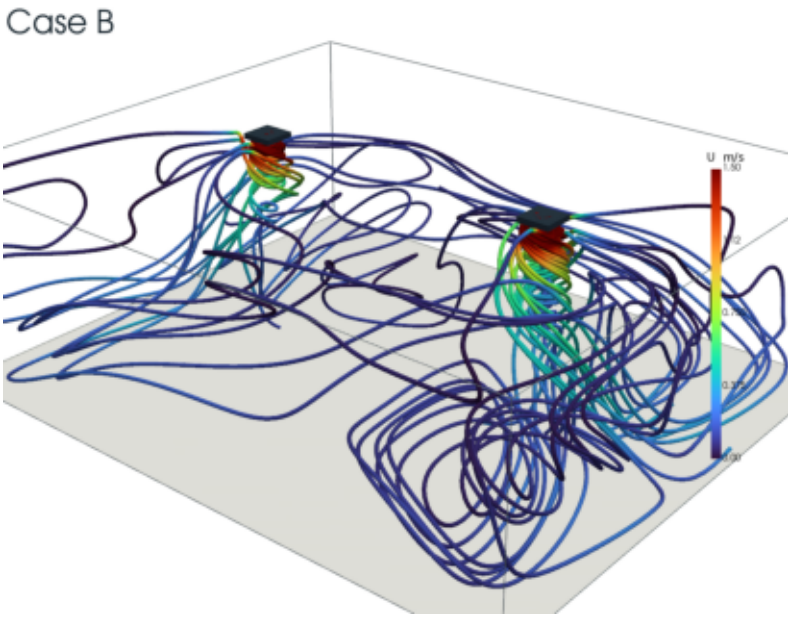
Flow structure — 3D streamlines from the 8 supply slots



Baseline 4-way
mean 24.4 °C
exit 19.2 °C
coverage 89%



Steep
mean 27.1 °C
exit 17.7 °C
coverage 68%



Swirl
mean 27.9 °C
exit 16.8 °C
coverage 4%

Methodology & caveats

MODEL OpenFOAM v2012 buoyantPimpleFoam (transient, compressible perfect gas), k-ω SST, 576,000 cells, 4-core. Hall 480 m³; two 0.6×0.6 m 4-way cassettes (0.4×0.4 central return + 4 supply slots) at (3,5) and (9,5).

CONTROLLED VARIABLE All patterns fix the vertical supply component at −2.83 m/s → identical inlet area 0.64 m², identical airflow 1.81 m³/s (13.6 ACH), identical mass flow 2.23 kg/s. Only the horizontal throw differs: baseline radial, swirl tangential (coherent CCW), steep reduced-throw. Mass conserved to ~1e-10.

LOAD / BC Supply 14 °C; adiabatic walls; 12 kW sensible load (~100 W/m²); uniform-24 °C start. Energy balance validated: baseline exit 19.2 °C vs 19.4 °C predicted by $Q = \dot{m} \cdot cp \cdot \Delta T$.

WHY THE MEAN DIFFERS (open system) The room exchanges heat with its surroundings through the return outflow. $dU/dt = Q - \dot{m} \cdot cp \cdot (T_{exit} - T_{supply})$. Q, ṁ, T_supply are identical, so the ONLY difference is the exit temperature, which the flow pattern sets. Lower exit T (short-circuit) → less heat carried out → warmer room.

CAVEATS Comparison is a matched snapshot at t = 350 s. Baseline is at steady state; swirl & steep are still approaching a hotter equilibrium (very long time-constant due to short-circuiting; not force-converged here). Single mesh (576k) — mesh-independence pending. RANS (modelled, not resolved, turbulence).

NEXT Mesh-independence on the baseline; an under-sized-airflow case (coverage vs ACH); optional full-asymptote runs for swirl/steep if the converged hot-equilibrium numbers are needed.